Weekly Lesson Plan – Week 1 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon | Week of: Aug 10, 2026 | Course: Engineering Design & Problem Solving (InSPIRESS)

TEKS / ELPS:

TEKS: §127.406(d)(8)(A) maintain records of the engineering design process; 127.15(d)(2) employability skills; §127.406(d)(4)(B) communicate findings and conclusions; (d)(3)(C) use mathematical calculations to assess quantitative relationships; §127.406 engineering design process - define a problem, generate ideas, plan a solution; §127.406 develop & refine a design; communicate it with digital tools; §127.406 present & evaluate a design solution
ELPS: c.1A, c.2D, c.3C, c.4G, c.5B

Aim of the Lesson / Objective: *Statement of what you want students to do (skill) and/or know (knowledge). Students will…*

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Students will set up a professional engineering design notebook to documentation standards (numbered, dated, permanent, witness-ready), sequence the seven stages of the formal engineering design process, distinguish design criteria from constraints, and write a first notebook entry that frames a candidate InSPIRESS payload as a measurable design problem.	Students will analyze technical communication using the claim–evidence–interpretation structure, apply significant-figure rules to measured quantities, convert and combine units by dimensional analysis with cancellation shown, and report a measured result with appropriate uncertainty.	Students will apply the engineering design process to define a real-world problem and begin designing a solution (Ask, Imagine, Plan).	Students will finish their final design, write a structured AI prompt, and generate + submit a product image.	Students will present their product - communicating the problem it solves, their final design, and how it works - and give supportive peer feedback.

Employed Language Domain(s) Objective: *How students achieve the objective – Listening, Speaking, Reading, Writing. Students will…*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.

Assessment of Content Objective: *Instrument used to determine mastery. Ex: Exit Ticket.*

Students will be assessed by written, group, or oral responses throughout the class period.

Set (Bell Work): *A motivating action that engages students in the objective – question, fact, activity. Essential Question:*

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
In one sentence, what separates an engineer from someone who just builds things?	Think of the best presentation you have ever seen. What made it good?	What is the purpose of the engineering design cycle?	Watch: how a product improves, version to version	Write your script before you present

Methodology / Steps: *What it takes to teach the objective so 100% can pass the assessment. Teacher will… Teams will… Individuals will…*

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Step 1: Bell ringer / essential question. Step 2: Bell-work: In your notebook, write one sentence on what separates *engineering* from *building*. We'll test it against the professional definition. (6 min) Step 3: The engineering design process: seven stages, and why documentation is 'stage zero' (10 min) Step 4: The language of a real problem: criteria vs. constraints, framed around the InSPIRESS payload challenge (10 min) Step 5: Build your design notebook to professional standard + write your first framed-problem entry (13 min) Step 6: Exit ticket: sort four statements into criteria/constraint and write one measurable success criterion (6 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: Write the best presentation you've ever seen and one concrete reason it worked. (5 min) Step 3: Watch a working engineer present, then name the *claim → evidence → interpretation* moves she makes (8 min) Step 4: The unit of a technical argument: claim → evidence → interpretation (7 min) Step 5: Significant figures, units & dimensional analysis — I do / we do / you do (13 min) Step 6: Measurement uncertainty: precision vs. accuracy and how to report it (8 min) Step 7: Exit ticket (4 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: purpose of the engineering design cycle (5 min) Step 3: Present: the 5-step engineering design process (16 min) Step 4: Pick a real problem (list 3, choose 1) (7 min) Step 5: Start the design sheet: Ask -> Imagine (4 idea sketches) -> Plan (Design #1 + materials) (17 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work + video (start 1:28): name 2 improvements and why each matters (8 min) Step 3: Model the notebook page (prompt on top, sketch on bottom) + copy the prompt (10 min) Step 4: Finish final design, then generate the product image in Gemini (20 min) Step 5: Submit the image to Google Classroom (7 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: write your 4-sentence presentation script (5 min) Step 3: Model: the four things every presenter says + how to be a great audience (8 min) Step 4: Presentations: each student shows their AI product image + notebook page (~1 min each) (32 min) Step 5: Exit ticket: one thing you would design next (5 min)

Resources: *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.

Re-Teaching Activities: *What you prepare if students do not master the objective.*

Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.

Reflection: *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state what they accomplished and what worked best.

Study Skills: *Skills implemented by students – note-taking, summarizing, teamwork, etc.*

Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.
SPED & 504 Accommodations / Modifications: *Codes.*
Per Individualized Education Plan (IEP) / 504 plan.
ELL Accommodations: *Codes.*
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).